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Windows PowerShell x Windows PowerShell x + v
=====
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG> notepad app\main.py
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG> Select-String -Path app\main.py -Pat
tern "retriever|as_retriever|similarity_search|Chroma|collection_name" -Context 3,3

app\main.py:6: ↓
app\main.py:7:LangGraph
app\main.py:8: ↓
> app\main.py:9:ChromaDB Retrieval
app\main.py:10: ↓
app\main.py:11:Ollama LLM
app\main.py:12: ↓
app\main.py:20:import time
app\main.py:21:
app\main.py:22:import mlflow
> app\main.py:23:from langchain_chroma import Chroma
app\main.py:24:from langchain_ollama import ChatOllama, OllamaEmbeddings
app\main.py:25:from langgraph.graph import END, START, StateGraph
app\main.py:26:
app\main.py:35:
app\main.py:36:MLFLOW_DB = BASE_DIR / "mlflow.db"
app\main.py:37:
> app\main.py:38:CHROMA_PATH = BASE_DIR / "chroma_db"
app\main.py:39:
app\main.py:40:OLLAMA_MODEL = "llama3.2:3b"
app\main.py:41:
app\main.py:67:
app\main.py:68:
app\main.py:69:# =====
> app\main.py:70:# CHROMADB VECTOR STORE
app\main.py:71:# =====
app\main.py:72:
> app\main.py:73:vectorstore = Chroma(
> app\main.py:74:     persist_directory=str(CHROMA_PATH),
app\main.py:75:     embedding_function=embeddings,
app\main.py:76:)
```

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app\main.py:78:
app\main.py:79:# =====
> app\main.py:80:# RETRIEVER
app\main.py:81:# =====
app\main.py:82:
> app\main.py:83:retriever = vectorstore.as_retriever(
app\main.py:84:     search_kwargs={"k": RETRIEVAL_K}
app\main.py:85:)
app\main.py:86:
app\main.py:111:
app\main.py:112:def retrieve(state: RAGState) -> RAGState:
app\main.py:113:    """
> app\main.py:114:    Retrieve relevant documents from ChromaDB.
app\main.py:115:    """
app\main.py:116:
> app\main.py:117:    documents = retriever.invoke(
app\main.py:118:        state["question"]
app\main.py:119:    )
app\main.py:120:
app\main.py:252:
app\main.py:253:    mlflow.log_param(
app\main.py:254:        "vector_database",
> app\main.py:255:        "ChromaDB",
app\main.py:256:    )
app\main.py:257:
app\main.py:258:    mlflow.log_param(

(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG> notepad app\main.py
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG> python app\main.py

=====
W11D5 - TRACKED AND EVALUATED RAG
=====

Enter your healthcare question: What are the common symptoms of diabetes?
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```
Windows PowerShell
=====
W11D5 TRACKED RAG PIPELINE
=====

Question:
What are the common symptoms of diabetes?

Retrieved Context:
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[1] Diabetes is a chronic condition in which blood glucose levels remain higher than normal. Type 1 diabetes occurs when the immune s
ystem attacks insulin-producing cells in the pancreas. Type 2 diabetes occurs when the body does not use insulin effectively or does
not produce enough insulin. Common symptoms of diabetes can include increased thirst, frequent urination, increased hunger, fatigue,
and blurred vision. Diabetes can be managed through healthy eating, physical activity, medication when prescribed, and regular monito
ring of blood glucose.

[2] 1. Diabetes

[3] Influenza, commonly called the flu, is a contagious respiratory illness caused by influenza viruses. Common symptoms can include
fever, cough, sore throat, muscle aches, headache, chills, and fatigue. Influenza can spread through respiratory droplets and close c
ontact. Vaccination and good hygiene practices can help reduce the risk of infection.

Answer:
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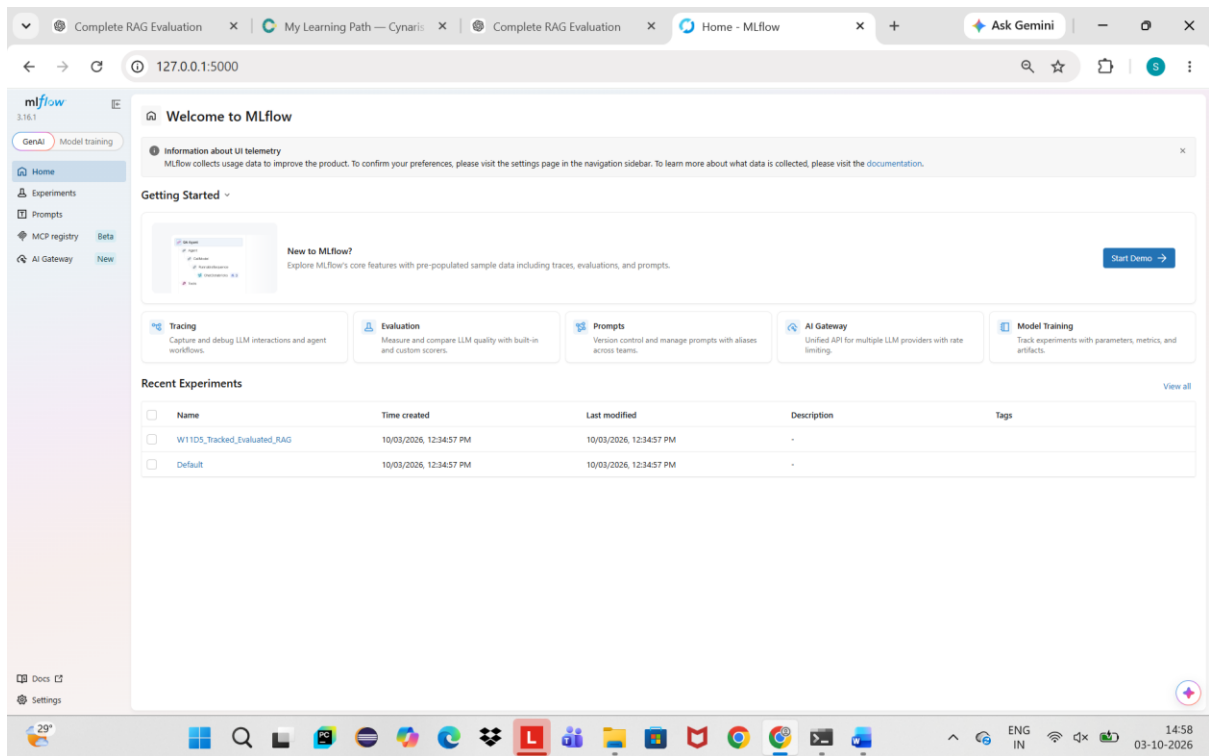
The common symptoms of diabetes include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision.

Execution time: 19.90s

Retrieved documents: 3

MLflow experiment: W11D5_Tracked_Evaluated_RAG
MLflow run ID: 7dd6149d3f634f86bca8e50f7ffce058
=====
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG> mlflow ui --backend-store-uri sqlite
:///mlflow.db
Registry store URI not provided. Using backend store URI.
```

```
Windows PowerShell
2026/10/03 13:58:20 INFO: Application startup complete.
2026/10/03 13:58:23 INFO: 127.0.0.1:63338 - "GET / HTTP/1.1" 200 OK
2026/10/03 13:58:23 INFO: 127.0.0.1:58103 - "GET /static-files/manifest.json HTTP/1.1" 200 OK
2026/10/03 13:58:24 INFO: Waiting for child process [16628]
2026/10/03 13:58:24 INFO: Child process [16628] died
2026/10/03 13:58:24 INFO: 127.0.0.1:63338 - "GET /static-files/static/js/main.32eac661.js HTTP/1.1" 200 OK
2026/10/03 13:58:24 INFO: 127.0.0.1:53842 - "GET /static-files/static/css/main.b0f58ef9.css HTTP/1.1" 200 OK
2026/10/03 13:58:28 INFO: 127.0.0.1:65178 - "GET /static-files/static/js/main.32eac661.js HTTP/1.1" 200 OK
2026/10/03 13:58:29 INFO: 127.0.0.1:54027 - "GET /static-files/Favicon.ico HTTP/1.1" 200 OK
2026/10/03 13:58:29 INFO: 127.0.0.1:63338 - "GET /static-files/TelemetryLogger.telemetry-worker.d0f8ea2c83c08b63528f.worker.js HT
TP/1.1" 200 OK
2026/10/03 13:58:29 INFO: 127.0.0.1:60308 - "GET /static-files/TelemetryLogger.telemetry-worker.d0f8ea2c83c08b63528f.worker.js HT
TP/1.1" 200 OK
2026/10/03 13:58:31 INFO: 127.0.0.1:53842 - "GET /ajax-api/3.0/mlflow/server-info HTTP/1.1" 200 OK
2026/10/03 13:58:32 INFO: 127.0.0.1:53842 - "GET /static-files/static/js/4205.9e750e7b.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:32 INFO: 127.0.0.1:60308 - "GET /static-files/static/js/4205.9e750e7b.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:32 INFO: 127.0.0.1:53842 - "GET /ajax-api/2.0/mlflow/users/current HTTP/1.1" 404 Not Found
2026/10/03 13:58:32 INFO: 127.0.0.1:54027 - "GET /ajax-api/3.0/mlflow/assistant/config HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:54027 - "GET /static-files/static/js/3877.386ccaf4.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:63338 - "GET /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:54027 - "GET /static-files/static/js/7340.cdf02082.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:63338 - "GET /static-files/static/js/5027.a66ffe35.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:59912 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:53842 - "GET /static-files/static/css/5027.26533251.chunk.css HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:53842 - "GET /ajax-api/2.0/mlflow/experiments/search?max_results=5&order_by=last_update_time+
DESC HTTP/1.1" 200 OK
2026/10/03 13:58:33 INFO: 127.0.0.1:59912 - "GET /static-files/static/media/demo-tracing-screenshot.f95794fa205aa8e6c386.png HTTP
/1.1" 200 OK
2026/10/03 13:58:34 INFO: 127.0.0.1:60308 - "GET /static-files/static/js/3877.386ccaf4.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:34 INFO: 127.0.0.1:55581 - "GET /static-files/static/js/7340.cdf02082.chunk.js HTTP/1.1" 200 OK
2026/10/03 13:58:34 INFO: 127.0.0.1:57912 - "GET /static-files/static/js/5027.a66ffe35.chunk.js HTTP/1.1" 200 OK
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG\.venv\Lib\site-packages\mlflow\server\fastapi_ap
p.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer t
o https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/03 13:58:49 INFO: Started server process [24376]
2026/10/03 13:58:49 INFO: Waiting for application startup.
```



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Windows PowerShell
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG\app\evaluate.py:20: DeprecationWarning: Importing
faithfulness from 'ragas.metrics' is deprecated and will be removed in v1.0. Please use 'ragas.metrics.collections' instead. Exampl
e: from ragas.metrics.collections import faithfulness
from ragas.metrics import (

=====
W11D5 RAGAS EVALUATION DATA
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Question: What are the common symptoms of diabetes?
Answer: The common symptoms of diabetes include increased thirst, frequent urination, increased hunger, fatigue, and blurred vision.
Retrieved documents: 3

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Question: What are the common symptoms of hypertension?
Answer: I don't have enough information in the provided context.
Retrieved documents: 3

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Question: What are the symptoms of asthma?
Answer: The symptoms of asthma include wheezing, coughing, chest tightness, and shortness of breath.
Retrieved documents: 3

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Question: What is a common cause of anemia?
Answer: A common cause of anemia is iron deficiency.
Retrieved documents: 3

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Question: Is the common cold caused by bacteria?
Answer: No, the common cold is not caused by bacteria.
Retrieved documents: 3

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Question: What is influenza?
Answer: Influenza is a contagious respiratory illness caused by influenza viruses.
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Answer: Influenza is a contagious respiratory illness caused by influenza viruses.
Retrieved documents: 3

-----
Question: What are some risk factors for heart disease?
Answer: According to the provided context, some risk factors for heart disease include:

1. High blood pressure
2. High cholesterol
3. Smoking
4. Diabetes
5. Physical inactivity
6. Obesity
7. Family history
Retrieved documents: 3

-----
Question: What is the purpose of preventive healthcare?
Answer: I don't have enough information in the provided context to answer the question about the purpose of preventive healthcare. The context only provides information about the measures included in preventive healthcare, but not its purpose.
Retrieved documents: 3

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STARTING RAGAS EVALUATION
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Evaluation LLM: Ollama
LLM model: llama3.2:3b

C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG\app\evaluate.py:281: DeprecationWarning: LangchainLLMWrapper is deprecated and will be removed in a future version. Use llm_factory instead: from openai import OpenAI; from ragas.llms import llm_factory; llm = llm_factory('gpt-4o-mini', client=OpenAI(api_key='...'))
  ragas_llm = LangchainLLMWrapper(
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG\app\evaluate.py:285: DeprecationWarning: LangchainEmbeddingsWrapper is deprecated and will be removed in a future version. Use the modern embedding providers instead: embedding_factory('openai', model='text-embedding-3-small', client=openai_client) or from ragas.embeddings import OpenAIEmbeddings, GoogleEmbeddings, HuggingFaceEmbeddings
```

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ragas_embeddings = LangchainEmbeddingsWrapper(
Evaluating: 0%| 0/32 [00:00<?, ?it/s]E
Exception raised in Job[0]: TimeoutError()
Evaluating: 3%| 1/32 [03:00<1:33:00, 180.02s/it]E
Exception raised in Job[1]: TimeoutError()
Exception raised in Job[6]: TimeoutError()
Exception raised in Job[3]: TimeoutError()
Exception raised in Job[7]: TimeoutError()
Exception raised in Job[2]: TimeoutError()
Exception raised in Job[4]: TimeoutError()
Exception raised in Job[5]: TimeoutError()
Exception raised in Job[14]: TimeoutError()
Exception raised in Job[12]: TimeoutError()
Exception raised in Job[13]: TimeoutError()
Exception raised in Job[10]: TimeoutError()
Exception raised in Job[8]: TimeoutError()
Exception raised in Job[11]: TimeoutError()
Exception raised in Job[9]: TimeoutError()
Exception raised in Job[15]: TimeoutError()
Evaluating: 59%| 19/32 [05:44<04:00, 18.52s/it]E
Exception raised in Job[16]: TimeoutError()
Exception raised in Job[18]: TimeoutError()
Exception raised in Job[22]: TimeoutError()
Exception raised in Job[20]: TimeoutError()
Exception raised in Job[23]: TimeoutError()
Exception raised in Job[26]: TimeoutError()
Exception raised in Job[28]: TimeoutError()
Exception raised in Job[24]: TimeoutError()
Exception raised in Job[25]: TimeoutError()
Exception raised in Job[27]: TimeoutError()
Evaluating: 78%| 25/32 [06:00<01:10, 10.10s/it]E
Exception raised in Job[29]: TimeoutError()
Exception raised in Job[30]: TimeoutError()
Exception raised in Job[31]: TimeoutError()
Evaluating: 100%| 32/32 [06:00<00:00, 11.25s/it]E

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Windows PowerShell x Windows PowerShell x + v
Exception raised in Job[10]: TimeoutError()
Exception raised in Job[8]: TimeoutError()
Exception raised in Job[11]: TimeoutError()
Exception raised in Job[9]: TimeoutError()
Exception raised in Job[15]: TimeoutError()
Evaluating: 59% | 19/32 [05:44<04:00, 18.52s/it]E
Exception raised in Job[16]: TimeoutError()
Exception raised in Job[18]: TimeoutError()
Exception raised in Job[22]: TimeoutError()
Exception raised in Job[20]: TimeoutError()
Exception raised in Job[23]: TimeoutError()
Exception raised in Job[26]: TimeoutError()
Exception raised in Job[28]: TimeoutError()
Exception raised in Job[24]: TimeoutError()
Exception raised in Job[25]: TimeoutError()
Exception raised in Job[27]: TimeoutError()
Evaluating: 78% | 25/32 [06:00<01:10, 10.10s/it]E
Exception raised in Job[29]: TimeoutError()
Exception raised in Job[30]: TimeoutError()
Exception raised in Job[31]: TimeoutError()
Evaluating: 100% | 32/32 [06:00<00:00, 11.25s/it]

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W11D5 RAGAS EVALUATION RESULTS
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faithfulness: nan
answer_relevancy: 1.0000
context_precision: nan
context_recall: 0.5000

Execution time: 659.07s

MLflow experiment: W11D5_Tracked_Evaluated_RAG
MLflow run ID: 4db25372ba5f4d1f98995b76e03cf807
=====
(.venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W11D5_Tracked_Evaluated_RAG> |
```